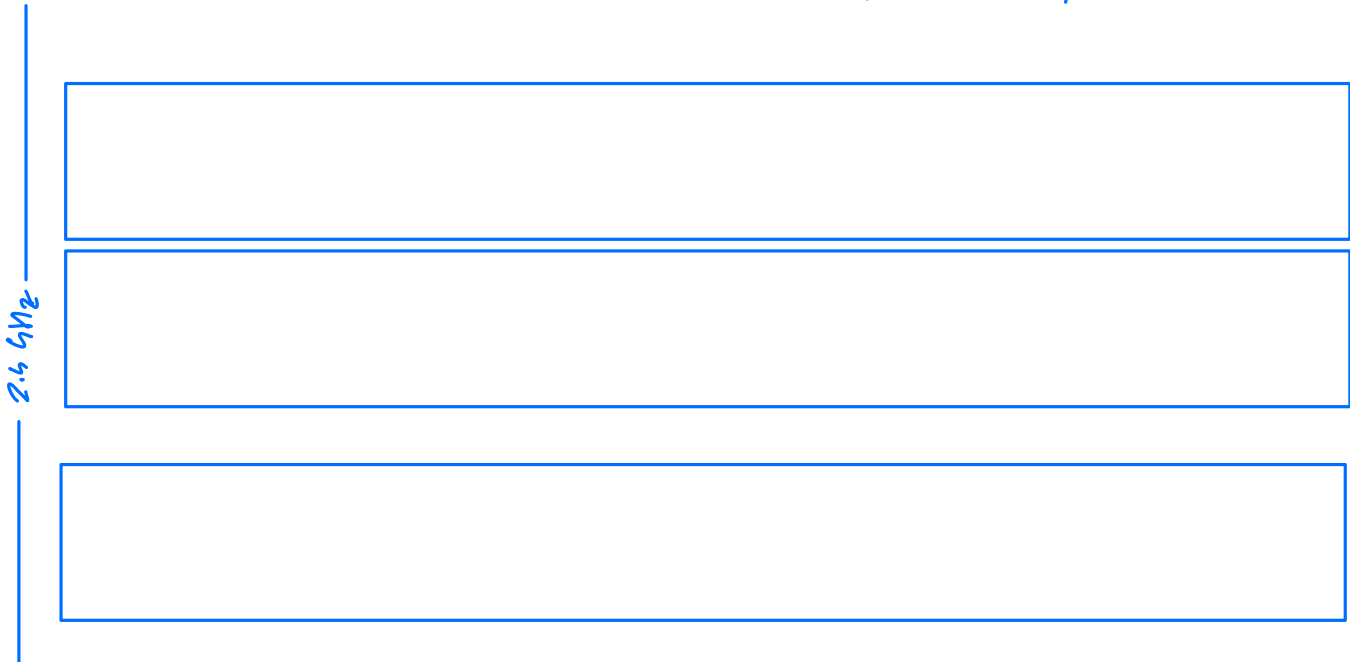


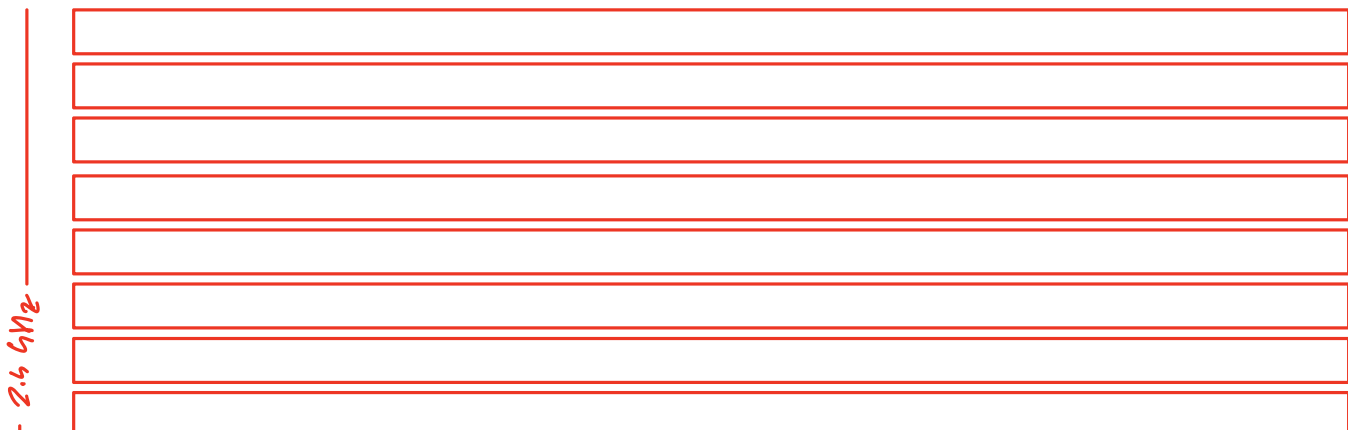
Bandwidth regulation with Zigbee by monitoring ED levels.

WiFi (IEEE 802.11) and Zigbee (IEEE 802.15.4) both use the 2.4 GHz band, unlicensed. WiFi has wider channels, while Zigbee has narrower ones.

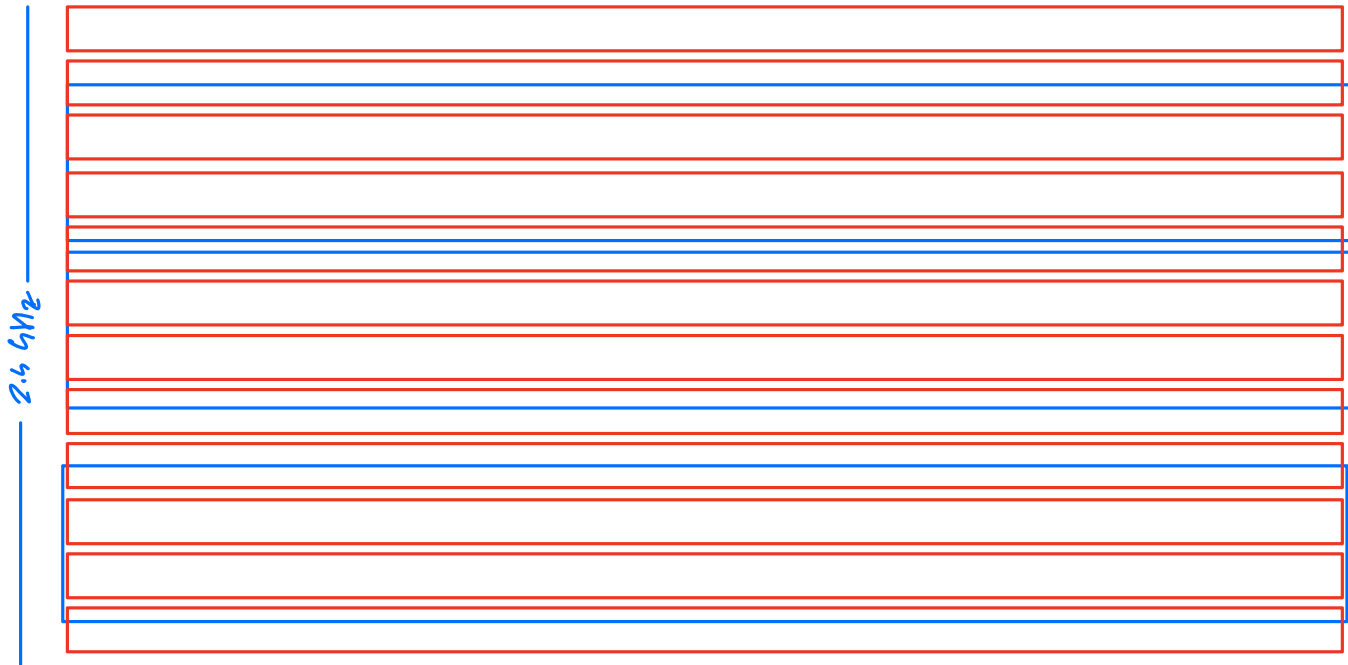
WiFi: ~ 6 channels, High power (20dBm+), 54+ Mbps



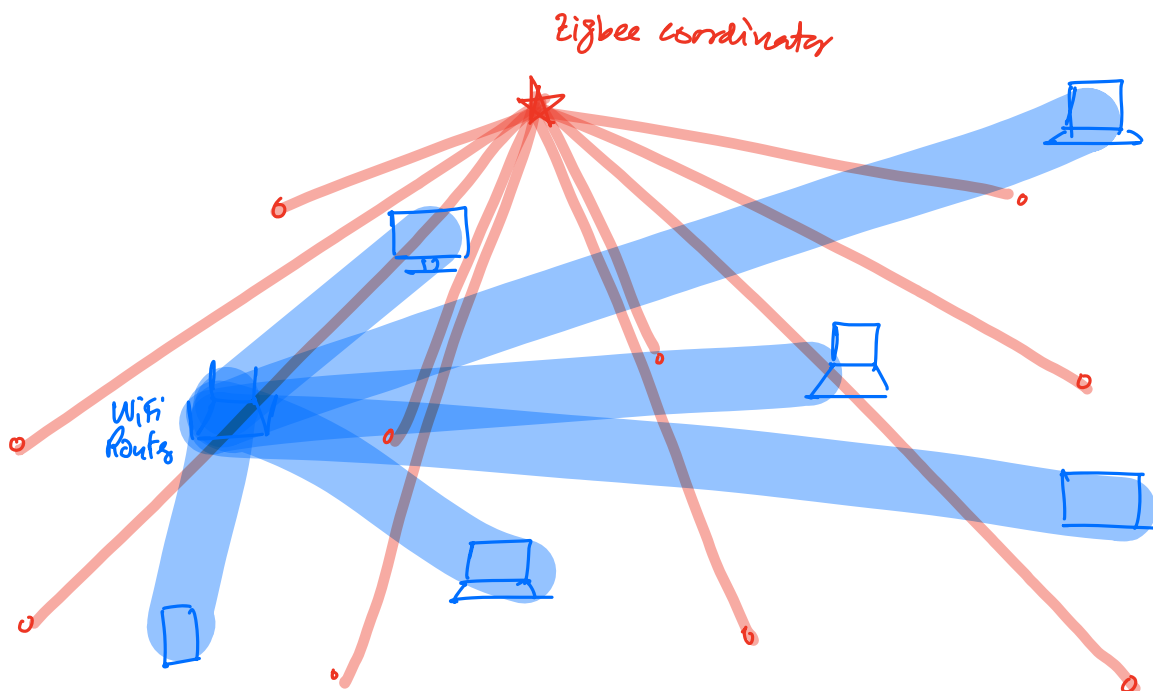
Zigbee: ~ 16 channels low power (~ 0 dBm), ~ 250 kbps



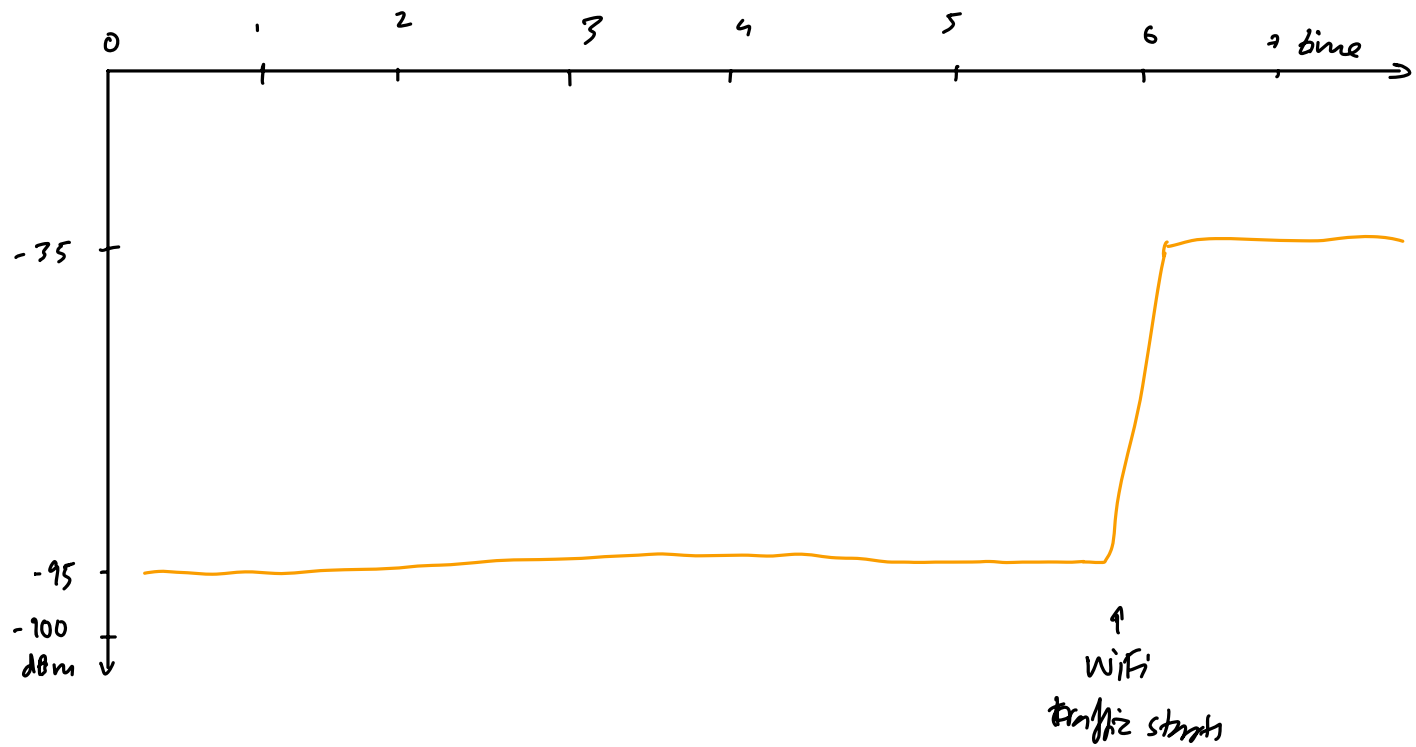
Both: (interference)



Energy Detection:



WiFi → high power traffic Tiny Zigbee packets get drowned and destroyed.



Our simulated system takes 500 ms to switch Zigbee devices from channel 12 (noisy) to channel 25 (at the high freq. gap where WiFi channels don't reach).

